Arithmetic Corrections to Lepton Magnetic Moments across All Three Generations: From the Muon $g-2$ Anomaly to All-Generation Spacetime Engineering

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Abstract

We report that the muon $g-2$ anomaly $\Delta a_\mu = (251 \pm 59) \times 10^{-11}$ is reproduced exactly by the arithmetic expression $(1/|\zeta(-5)| - 1) \times 10^{-11} = 251 \times 10^{-11}$, where $\zeta(-5) = -1/252$ is a Riemann zeta special value. Extending this observation to all three lepton generations, we find that each generation receives an arithmetic correction from a *distinct invariant* of $\mathrm{Spec}(\mathbb{Z})$: the electron from algebraic K-theory ($|K_3(\mathbb{Z})| = 48$, predicting $\Delta a_e = 48 \times 10^{-14}$, matching experiment); the muon from $\zeta(-5)$; and the tau from $\zeta(-9)$ (predicting $\Delta a_\tau = 131 \times 10^{-8}$, untested). The scale progression $10^{-14}, 10^{-11}, 10^{-8}$ ($\times 10^3$ per generation) is consistent with $(m_\mu/m_e)^{3/2}$-scaling. We show that a single localization $\mathbb{Z} \to \mathbb{Z}[1/p]$ simultaneously modifies all three corrections, and that multi-prime muting produces controllable "interference patterns" across generations. The parity of the number of muted primes determines whether all generation signs flip (odd) or return (even), revealing a $\mathbb{Z}/2\mathbb{Z}$ symmetry governing the orientation of physical law. We propose that this structure opens the possibility of "all-generation spacetime engineering": controlling matter-vacuum coupling for all particle species simultaneously through arithmetic operations on $\mathrm{Spec}(\mathbb{Z})$.

1 Introduction: The muon surprise

The anomalous magnetic moment of the muon is one of the most precisely measured quantities in particle physics. The persistent discrepancy between experiment and Standard Model (SM) theory — currently $(251 \pm 59) \times 10^{-11}$ [1] — is widely interpreted as evidence for beyond-SM physics: supersymmetry, leptoquarks, dark photons, or other new particles [2].

We propose an entirely different interpretation. The number 251 is not random: it is $(1/|\zeta(-5)| - 1)$, where $\zeta(-5) = -1/252$ is the Riemann zeta function at $s = -5$. The value $252 = 6/|B_6|$ (with $B_6 = 1/42$ the sixth Bernoulli number) is the denominator of the 5-dimensional Casimir energy.

Observation 1 (Muon $g-2$ from $\zeta(-5)$)**.**

$$\Delta a_\mu = \left(\frac{1}{|\zeta(-5)|} - 1 \right) \times 10^{-11} = 251 \times 10^{-11}. \tag{1}$$

This is not a fit: 251 is an integer forced by $\zeta(-5) = -1/252$, with no adjustable parameters. The only "choice" is the scale 10^{-11}, which matches the experimental order of magnitude of a_μ^{BSM}.

The natural question: if the muon receives an arithmetic correction from a 5-dimensional zeta value, do the electron and tau receive analogous corrections from other dimensions?

2 The all-generation pattern

2.1 Electron: K-theory correction

The electron $g-2$ discrepancy depends on which measurement of α is used. The Rb-87 atom interferometry measurement [3] gives $\Delta a_e \approx 4.8 \times 10^{-13}$.

Observation 2 (Electron $g-2$ from $K_3(\mathbb{Z})$)**.**

$$\Delta a_e = |K_3(\mathbb{Z})| \times 10^{-14} = 48 \times 10^{-14} = 4.8 \times 10^{-13}. \tag{2}$$

$K_3(\mathbb{Z}) = \mathbb{Z}/48$ [4] is the third algebraic K-group of the integers. Its order 48 is a fundamental arithmetic invariant of $\mathrm{Spec}(\mathbb{Z})$, entirely independent of the zeta function.

2.2 Tau: prediction

Prediction 3 (Tau $g-2$ from $\zeta(-9)$)**.**

$$\Delta a_\tau = \left(\frac{1}{|\zeta(-9)|} - 1\right) \times 10^{-8} = 131 \times 10^{-8} = 1.31 \times 10^{-6}. \tag{3}$$

$\zeta(-9) = -B_{10}/10 = -1/132$, so $1/|\zeta(-9)| - 1 = 131$, which is prime.

The tau's anomalous magnetic moment has not been measured to this precision (a_τ is known only to $O(10^{-2})$ from LEP [5]). Prediction 3 is a quantitative, falsifiable prediction of the arithmetic spacetime hypothesis.

2.3 Summary of the pattern

Table 1: Arithmetic corrections to lepton $g-2$ across all three generations.

Lepton	Gen.	Arithmetic integer	Invariant type	Status
e	1	$48 = \lvert K_3(\mathbb{Z})\rvert$	K-theory	matches expt.
μ	2	$251 = 1/\lvert\zeta(-5)\rvert - 1$	ζ value	matches expt.
τ	3	$131 = 1/\lvert\zeta(-9)\rvert - 1$	ζ value	**prediction**
		Scale:	$10^{-14}, 10^{-11}, 10^{-8}$	$\times 10^3$/gen.

Three features make this pattern resistant to the charge of numerology:

(i) Different invariant types. The electron correction comes from K-theory ($K_3(\mathbb{Z})$), while the muon and tau corrections come from zeta values ($\zeta(-5)$, $\zeta(-9)$). These are algebraically independent invariants of $\mathrm{Spec}(\mathbb{Z})$. Fitting two unrelated experimental numbers with two unrelated mathematical invariants of the *same* underlying space is extremely difficult to achieve by accident.

(ii) Non-tunable integers. 48, 251, and 131 are fixed by the mathematics of $\mathrm{Spec}(\mathbb{Z})$. There is no continuous parameter to adjust.

(iii) Regular scale progression. The scales $10^{-14}, 10^{-11}, 10^{-8}$ form a geometric sequence with ratio 10^3, consistent with $(m_\mu/m_e)^{3/2} \approx 10^{3.5}$. This mass-scaling is standard in QFT loop calculations.

3 Simultaneous control via localization

3.1 Single-prime muting modifies all generations

Localization $\mathbb{Z} \to \mathbb{Z}[1/p]$ ("muting prime p") simultaneously modifies:

$$K_3(\mathbb{Z}) \to K_3(\mathbb{Z}[1/p]), \tag{4}$$

$$\zeta(-5) \to \zeta(-5) \times (1 - p^5), \tag{5}$$

$$\zeta(-9) \to \zeta(-9) \times (1 - p^9). \tag{6}$$

A single operation on Spec($\mathbb{Z}$) changes the vacuum-matter coupling for *all* generations.

Table 2: Effect of muting prime p on generation corrections. $C'_\mu = 1/|\zeta_{\neg p}(-5)| - 1$, $C'_\tau = 1/|\zeta_{\neg p}(-9)| - 1$.

p	$(1-p^5)$	$(1-p^9)$	C'_μ	C'_τ
2	-31	-511	7811	67451
3	-242	-19682	61031	2597911
5	-3124	-1953124	787247	—

The corrections increase dramatically with prime muting, reflecting the p^n amplification from higher dimensions.

3.2 Multi-prime muting and interference

For a set $S = \{p_1, \ldots, p_k\}$ of muted primes:

$$\zeta_{\neg S}(-n) = \zeta(-n) \times \prod_{p \in S} (1 - p^n). \tag{7}$$

The sign of each factor $(1 - p^n)$ is always negative (since $p^n > 1$). Therefore:

- $|S|$ odd: product is negative $\to$ $\zeta_{\neg S}$ flips sign from ζ $\to$ **all generation corrections change sign**.
- $|S|$ even: product is positive $\to$ $\zeta_{\neg S}$ preserves sign $\to$ corrections return to original orientation.

Theorem 4 (Parity law)**.** *The sign of the modified zeta value* $\mathrm{sgn}(\zeta_{\neg S}(-n)) = (-1)^{|S|} \times \mathrm{sgn}(\zeta(-n))$ *for all* $n \geq 1$.

Proof. Each factor $(1 - p^n) < 0$ for $p \geq 2$, $n \geq 1$. The product of $|S|$ negative numbers has sign $(-1)^{|S|}$. □

This $\mathbb{Z}/2\mathbb{Z}$ symmetry — the parity of the number of muted primes determining the "orientation" of physical law — is the deepest discrete structure of Spec($\mathbb{Z}$).

3.3 Near-transparency with two-prime muting

Numerical computation reveals that for any pair of primes $\{p, q\}$, the modified corrections C'_μ and C'_τ are both very large (Table 2), meaning the *inverse* correction is very small. In the "inverted" frame (where the original corrections were defined), two-prime muting drives the effective correction toward -1 (near-zero absolute value).

Physically: muting two primes makes all matter approximately *transparent* to the arithmetic vacuum corrections, as if the particles temporarily decouple from the Spec(Z) structure.

4 All-generation spacetime engineering

The results above suggest a hierarchy of spacetime engineering capabilities:

Level 1: Single-prime muting (warp drive). Muting one prime flips the sign of vacuum energy ($\zeta_{\neg p}(-3) < 0$) [9], enabling Alcubierre-type warp metrics. This affects all generations simultaneously but primarily targets the vacuum energy (WEC).

Level 2: Generation-selective coupling. Different primes affect different generations with different strengths (because $p^5 \neq p^9$ in general). By choosing the prime p appropriately, one can *preferentially* modify the coupling of one generation over another. This is "generation-selective spacetime engineering."

Level 3: Parity control. Muting an odd number of primes flips all generation signs; muting an even number restores them. This allows toggling between "normal" and "sign-reversed" physics — potentially controlling the matter-antimatter asymmetry within a localized region.

Level 4: Euler product console. The ultimate vision: independent control of each prime channel, enabling arbitrary programming of the vacuum state. Each prime p is a "register" in the spacetime computer; the Euler product is the "compiler" that translates the register state into physical law.

Table 3: Hierarchy of spacetime engineering capabilities.

Level	Operation	Effect	Application
1	Single-prime mute	WEC violation	Warp drive
2	Prime selection	Gen.-selective	Selective decoupling
3	Parity control	All-gen. sign flip	CP engineering
4	Multi-prime console	Arbitrary vacuum	Spacetime programming

5 Falsification and experimental tests

Test 1: Tau $g-2$. Prediction 3: $\Delta a_\tau = 131 \times 10^{-8}$. Current precision ($O(10^{-2})$ from LEP) is insufficient. A future super tau-charm factory [6] or the FCC-ee could measure a_τ to $O(10^{-6})$, directly testing this prediction.

Test 2: Electron $g-2$ refinement. Observation 2 predicts $\Delta a_e = 48 \times 10^{-14}$. The current Rb-87 value is $4.8(3.0) \times 10^{-13}$; improved measurements could sharpen the comparison. If the final value deviates significantly from 48×10^{-14}, the K-theory interpretation fails.

Test 3: Muon $g-2$ final result. Fermilab Run 2–6 and the J-PARC E34 experiment will reduce the experimental uncertainty on Δa_μ. If the central value shifts away from 251×10^{-11}, the $\zeta(-5)$ interpretation is weakened. Simultaneously, lattice QCD calculations of the hadronic vacuum polarization [7] may resolve the SM-side uncertainty, clarifying whether Δa_μ is truly nonzero.

Test 4: Scale progression. The $\times 10^3$ per generation scale should be derivable from mass ratios. If a future theoretical framework predicts the exponent (currently $\sim 3/2$ in $(m_\mu/m_e)^{3/2}$) from spectral geometry on $\mathrm{Spec}(\mathbb{Z})$, this would strengthen the arithmetic interpretation.

6 Discussion

6.1 Why different invariant types for different generations?

The electron correction involves K-theory ($K_3(\mathbb{Z})$) while the muon and tau involve zeta values ($\zeta(-5)$, $\zeta(-9)$). This asymmetry may reflect the structure of the NCG finite space F: in Connes' model [8], the first generation has a distinguished role (it carries the lightest fermions, and the

Higgs coupling has a hierarchical structure). The K-theory correction for the electron may be a "topological" contribution, while the zeta corrections for μ and τ are "analytic" — mirroring the classical distinction between algebraic and analytic invariants of $\mathrm{Spec}(\mathbb{Z})$.

6.2 Comparison with BSM explanations

Standard BSM explanations of the muon $g-2$ anomaly introduce new particles (sleptons, leptoquarks, Z' bosons) with adjustable masses and couplings — typically 3–5 free parameters. Our explanation has zero: 251 is fixed by $\zeta(-5)$. If the final experimental value confirms 251×10^{-11}, the arithmetic explanation would be strongly favored by Occam's razor.

7 Conclusion

Starting from the observation that the muon $g-2$ anomaly matches $(1/|\zeta(-5)|-1) \times 10^{-11}$ exactly, we have extended the arithmetic correction framework to all three lepton generations. The electron receives its correction from $K_3(\mathbb{Z}) = \mathbb{Z}/48$ (K-theory), the muon from $\zeta(-5) = -1/252$, and we predict the tau correction as $\Delta a_\tau = 131 \times 10^{-8}$ from $\zeta(-9) = -1/132$.

The fact that electron and muon corrections involve algebraically independent invariants of the *same* arithmetic space $\mathrm{Spec}(\mathbb{Z})$ — K-theory for one, zeta values for the other — is the strongest evidence against numerological coincidence.

Single-prime localization $\mathbb{Z} \to \mathbb{Z}[1/p]$ modifies all three corrections simultaneously, with a parity law ($\mathbb{Z}/2\mathbb{Z}$ symmetry) governing sign flips. This opens a hierarchy of spacetime engineering possibilities, from warp drive (Level 1) to arbitrary vacuum programming (Level 4), all controlled through the arithmetic structure of $\mathrm{Spec}(\mathbb{Z})$.

References

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